

# Daniel-John DIALA

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## EDUCATION

**Monmouth University**, West Long Branch, NJ  
Bachelor of Science in Software Engineering · Honors School  
**Honors:** Dean's List (Fall 2025, Spring 2026); Academic Excellence Scholarship; Shirley Family Scholarship  
**Coursework:** Data Structures & Algorithms, Discrete Mathematics, Computer Architecture, Calculus I & II

Expected May 2029  
**GPA:** 4.0/4.0

## EXPERIENCE

**Undergraduate Research Assistant** | Monmouth University  
• Modernized GArEl, an OCaml bidirectional type checker (from Qu et al., ICFP 2019) that analyzes program cost statically  
• Fixed drift in a 36-program test corpus, stress-tested the checker, and developed one of two independent solution paths  
• Currently evaluating CHC- and SMT-solver-based alternatives to scale the analysis  
• Built an educational web app, hosted on AWS, that visualizes constraint generation to teach bidirectional type checking

May 2026 – Present

**AI Literacy Teaching Assistant** | Monmouth University  
• Help students set up agentic AI systems and use AI tools effectively and ethically  
• Built a Next.js/TypeScript exam-prep web app with 180 practice questions and timed mock exams for Anthropic's Claude Certified Architect – Foundations exam, deployed on AWS EC2

Sep 2026 – Present

## PROJECTS

**Pulse-Net: Federated Pandemic Early-Warning System** | Python, PyTorch, FastAPI, PostgreSQL, React  
• Built a federated learning pipeline where 5 simulated hospitals train PyTorch models locally and send signed, differentially private ( $\epsilon = 1.0$ ) updates, merged with a norm-clipped coordinate-wise median to resist poisoning  
• Designed a transformer classifier with supervised contrastive loss that flags novel pathogens by embedding distance; 83.6% validation accuracy across 16 classes on synthetic data  
• Built encrypted aggregation with CKKS, Shamir k-of-n key escrow, zero-knowledge range proofs, and a Merkle audit log  
• Built a React dashboard with a live outbreak map, WebSocket alerts, and SEIR spread models; 1,500+ pytest tests run in CI

Jan 2026 – Present

**Sentiment Analysis Tool** | Python, spaCy, Flask, SQLite, Streamlit, Plotly  
• Built an NLP engine that classifies reviews as positive, negative, neutral, or mixed using a 6,800-term graded lexicon with domain variants, negation handling, and emoji sarcasm cues  
• Added aspect-based analysis and a Flask REST API (15+ endpoints) feeding a Streamlit trend dashboard  
• Cut runtime for 500 reviews from 17s to 5s (3.4×) via batched spaCy processing, lemma caching, and bulk SQLite inserts

Dec 2025 – Jan 2026

## SKILLS & CERTIFICATIONS

**Languages:** Python, Java, TypeScript, JavaScript, SQL, OCaml, HTML/CSS  
**ML & Data:** PyTorch, spaCy, NLP, federated learning, differential privacy, Plotly  
**Web & Infra:** FastAPI, Flask, React, Next.js, PostgreSQL, Redis, AWS (EC2), Docker, Git, GitHub Actions, pytest  
**Certification:** Claude Certified Architect – Foundations, Anthropic (Jul 2026)

## ACTIVITIES

**IEEE/ACM Student Chapter**, University & External Relations Committee Chair  
**CyberHawks:** National Cyber League (Spring 2026), team top 4%, individual top 14%

Mar 2026 – Present  
Sep 2025 – Present